

# EMILY YU

**Assistant Professor, Leiden Institute of Advanced Computer Science, Leiden University**

<https://emillyu.nl/> — Google Scholar — [Z.YU@LIACS.LEIDENUNIV.NL](mailto:Z.YU@LIACS.LEIDENUNIV.NL) — 0000-0002-4993-773X

Address: Gorlaeus Building, Einsteinweg 55, 2333 CC Leiden, Netherlands

## RESEARCH INTERESTS

---

Formal verification and automated reasoning, especially for hardware designs and AI-based systems;  
Runtime monitoring and control: ensuring safety and correctness of complex dynamical systems.

## EDUCATION

---

**Johannes Kepler University Linz, Austria** 2019 – 2023  
Ph.D. in Computer Science  
“mit Auszeichnung bestande” (degree with distinction)  
Thesis: *Certifying Hardware Model Checking*  
Supervisor: Prof. Armin Biere

**Imperial College London, United Kingdom** 2014 – 2018  
M.Eng in Computing, *First Class Honours*  
Integrated Master’s degree: 3-year Bachelor + 1-year Master  
Thesis: *Model Checking Temporal Epistemic Logic under Bounded Recall*  
Supervisor: Prof. Alessio Lomuscio

**Concord College, United Kingdom** 2012 – 2014  
GCE A-levels (High School)

## PROFESSIONAL EXPERIENCE

---

**Leiden University, Netherlands** Since Oct 2025  
Tenured Assistant Professor  
Leader of the Veritas Lab: Safe AI and Automated Reasoning.  
Leiden Institute of Advanced Computer Science

**Institute of Science and Technology Austria, Austria** Sept 2023 – Sept 2025  
Postdoctoral Researcher  
Henzinger Group: Design and Analysis of Concurrent and Embedded Systems  
Working with Prof. Thomas A. Henzinger on runtime monitoring and neural control verification.

**Woodpecker Technologies, Singapore** Aug 2022 – Jan 2023  
External Consultant  
Pre-silicon formal verification for chip designs.

**Johannes Kepler University Linz, Austria** Feb 2019 – Jun 2023  
Project Assistant  
Formal Models and Verification Group, lead by Prof. Armin Biere.

**Morgan Stanley, London, United Kingdom** April – Sept 2017  
Software Engineer  
Full-stack development during a 6-month industrial placement.

**BlackRock, London, United Kingdom** Jun – Aug 2016  
Software Engineer Intern  
Worked on internal web platform development.

**Imperial College London, United Kingdom**

2015 – 2016

Teaching Assistant

Part-time employment. Courses: Discrete Maths, Logic, and Reasoning about Programs.

**Imperial College London, United Kingdom**

Jul – Sept 2015

Research Assistant

20 hours per week. Group of Prof. Chris Hankin. Research on games and abstractions for cybersecurity.

**CAREER BREAKS**

---

Maternity leave 2021-2022, and 2026.

**HONORS & AWARDS**

---

**NWO Veni**, Principal Investigator

2026

*VAISS: Verified Analysis for Infinite-State Systems*

Dutch Research Council, Talent Programme

EUR 320,000 (14% acceptance rate)

**Distinguished Paper Award**, Int. Conf. on Computer Aided Verification (CAV) 2026

2026

★ CORE ranking: A\*

**FWF ESPRIT Fellowship**, Principal Investigator

2025

*RAVENS: Runtime Assurance and Verification of Neural Systems*

Austrian Science Fund FWF

Host Institute: Vienna University of Technology, Austria

EUR 346,505

**Distinguished Paper Award**, Int. Conf. on Computer Aided Verification (CAV) 2025

2025

★ CORE ranking: A\*

**Disruptive Idea Award**, Neuro-symbolic Systems (NeuS) Conference

2025

DARPA award, USD 100,000

**LANGUAGE PROFICIENCY**

---

English (fluent), Cantonese (fluent), German (B2), Mandarin (native)

**ACADEMIC SERVICES AND OUTREACH**

---

**Member of Program Committee**

- AAAI Conference on Artificial Intelligence (AAAI), 2027;
- SETSS 2026 Spring School on Engineering Trustworthy Software Systems, 2026;
- Int. Conference on Computer Design (ICCD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2026;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2026;
- Int. Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026);
- Int. Symposium of Electronics Design Automation (ISED), 2026;
- Post-AI Formal Methods Workshop, 2026;
- AAAI Conference on Artificial Intelligence (AAAI), 2026;
- European Conference on Artificial Intelligence (ECAI), 2025;
- Int. Joint Conference on Artificial Intelligence (IJCAI), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD), 2025;
- Int. Conference on Formal Methods in Computer-Aided Design (FMCAD) Student Forum, 2025;

- Int. Conference on integrated Formal Methods (iFM), 2024.

#### **Member of Artifact Evaluation Committee**

- Int. Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI), 2024;
- Int. Symposium on Formal Methods (FM), 2024;
- Int. Symposium on Automated Technology for Verification and Analysis (ATVA), 2024.

#### **Journal Referee**

- Science of Computer Programming, 2025.
- Journal of Artificial Intelligence Research (JAIR), 2025
- Formal Aspects of Computing Journal, 2025.

#### **Conference (sub-)Referee**

- IEEE Int. Symposium on Circuits and Systems (ISCAS) 2026;
- Int. Conference on Neuro-symbolic Systems (NeuS), 2025;
- Int. Conference on Automated Deduction (CADE), 2025.

#### **Scientific Referent for Funding Agencies**

- French National Research Agency (ANR), 2025.

#### **Event Organization**

- IPA Fall days 2026 (PhD school on Software Engineering and AI). Member of the organizing team.
- CAV 2021, Virtual. Member of the volunteering team.
- QONFEST 2020, Vienna, Austria. Member of the volunteering team.
- Kurt Goedel Conference 2019, Vienna, Austria. Member of the volunteering team.

#### **PhD Examination and Reviewing Committees**

- PhD thesis committee member of Annelot Bosman, Leiden University, 2026.

#### **Other Committees**

- Committee member of the Diversity Committee of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.
- Committee member of the LIAXX (women's network) of the Leiden Institute of Advanced Computer Science (LIACS), Leiden University, since 2025.

### **PHD/POSTDOC SUPERVISION**

---

Harshul Gupta (PhD student) at Leiden University, starting June 2026.

Yanni Dong (Postdoc) at Leiden University, May 2026 - Present.

### **STUDENT SUPERVISION**

---

Master's thesis supervisor at Leiden University for Guangyu Zhu, Feb 2026 - Present.  
Thesis title: *Signal First-Order Logic Neural Control*

Master's thesis supervisor at Leiden University for Zhengqi Lang, Oct 2025 - July 2026.  
Thesis title: *Robust Certificates for Neural Control*

Bachelor's thesis supervisor at Leiden University for Enzo Beekman, Feb - July 2026.  
Thesis title: *Certifying Constraints from Hardware Designs*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Spring 2025. Project topic: *Verification of Logic Gate Neural Networks*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Sergei Pankratov, Spring 2025. Project topic: *Certificate-based Neural Control*

PhD rotation supervisor at Institute of Science and Technology Austria (ISTA) for Fabian Kresse, Winter 2024. Project topic: *Predictive Runtime Monitoring for Complex Dynamical Systems*

## TEACHING EXPERIENCE

---

Lecturer, **Formal Methods for AI.** Since Autumn 2026

Master's course, 6 ECTS

Mandatory course for CS Master's students in Foundations of Computing

[This is a self-designed course by myself.](#)

Leiden University, Netherlands

Lecturer, **Software Engineering.** Since Spring 2026

Bachelor's course, 6 ECTS

Students complete real-world software projects in collaboration with LUdev, a student-run software company. This course also contains self-designed teaching materials.

Leiden University, Netherlands

Teaching Assistant, **Discrete Maths, Reasoning about Programs, Logic.** 2015 - 2016

Imperial College London, London, United Kingdom.

## RESEARCH VISITS & OTHERS

---

June-July 2024, Prof. Joost-Pieter Katoen, RWTH Aachen University, Aachen, Germany.

June-July 2024, Prof. Jürgen Giesl, RWTH Aachen University, Aachen, Germany.

Since 2023, core contributor to an Intel funded project on certifying hardware model checking; PI: Prof. Armin Biere.

Former member of the ERC project VAMOS (2023-2025); PI: Prof. Thomas A. Henzinger.

Core developer of certificate checkers used for hardware model checking competitions.

Former member of an EPSRC project titled Games and Abstractions (2015), under the supervision of Prof. Chris Hankin at Imperial College London.

## ACADEMIC REFERENCES

---

Prof. Thomas A. Henzinger (Institute of Science and Technology Austria) – [tah@ist.ac.at](mailto:tah@ist.ac.at)

Prof. Armin Biere (University of Freiburg, Germany) – [biere@cs.uni-freiburg.de](mailto:biere@cs.uni-freiburg.de)

Prof. Keijo Heljanko (University of Helsinki, Finland) – [keijo.heljanko@helsinki.fi](mailto:keijo.heljanko@helsinki.fi)

Prof. Martina Seidl (Johannes Kepler University Linz, Austria) – [martina.seidl@jku.at](mailto:martina.seidl@jku.at)